

Optical Fi-Wi QKD Exchange with Monolithic QKD Transmitter

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Abstract We show a handheld-inspired end-to-end key exchange over optical-wireless channel and up to 32-km fronthaul link, using a silicon QKD transmitter with on-chip light source operable up to 80°C and 29 WDM channels. We further prove co-existence with a 5-Gb/s Fi-Wi data link. ©2026 The Authors

Introduction

The dawning quantum age and its threat of quantum computation to current asymmetric data encryption has propelled the deployment of quantum key distribution (QKD) [1]. However, current rack-scale form-factors are incompatible with consumer devices, which call for a chip-scale solution [2]. Toward that, photonic chip integration has been shown on monolithic InP [3-6], hybrid InP/silicon [6-10] and InP/polymer [6, 11, 12] platforms. However, the integration in handheld devices [13-15] remains a niche topic due to complexity considerations and free-space optical coupling – in great contrast to the success of smartphone integration in the classical radio-frequency communication realm.

In this work we introduce our monolithic silicon QKD transmitter engine [16] to a hybrid optical fiber-wireless (Fi-Wi) key exchange. We achieve positive secure-key rates (SKR) over 30-cm optical-wireless access and up to 32-km fronthaul reach. We show colorless and high-temperature QKD operation over 29 WDM channels and up to 80°C. We further confirm a positive SKR in co-existence with 5-Gb/s Fi-Wi data transmission.

Commodity-QKD Exchange over Fi-Wi Access

Figure 1a presents the key exchange scenario between Alice's QKD transmitter and Bob's edge-cloud receiver situated in the access-metro aggregation segment. The transparent end-to-end exchange over wireless and wired domains involves an optical wireless communication (OWC) link to a field-distributed access terminal serving as optical hotspot (OHS). At this OHS a focal plane array (FPA) beamformer robustly connects a mobile Alice to a centralized Bob

through a fiber fronthaul – without the need for an intermediate trusted node. QKD then fills a key memory at the mobile device. Cost concerns are addressed by realizing Alice's QKD transmitter on an all-silicon basis. This monolithic integration approach [17] builds on a SiGe light source to enable smooth blending with the cost-efficiency of microelectronics. It mitigates (i) III-V materials for light generation, (ii) the complex assembly of multi-die modules, and (iii) hermetic packaging.

Experimental Fi-Wi QKD Exchange Link

Figure 1b presents the lab setup. Alice's QKD transmitter PIC implements a 4-state polarization encoded BB84 QKD protocol through two phase modulation stages (PM_{1,2}), a 2×2 multi-mode interference (MMI) coupler, and a 2D grating coupler (GC) as polarizing beam combiner [17]. It launches the encoded states in *H*, *V*, *R*, and *L* polarizations at a rate of 100 MHz. Pulse carving at 50% through direct modulation of the on-chip forward-biased lateral SiGe LED-like light source guarantees the security of the key exchange. The long-term stability of the on-chip SiGe light source is shown in Fig. 1c over more than 13 days. The peak power of -65.5 dBm during the first 11 days of single-mode fiber (SMF) coupling provides sufficient margin for QKD protocol encoding. The large swing of 8 dB is attributed to instabilities in passive fiber coupling due to temperature fluctuations: When instead coupling to 50 μm multi-mode fiber (MMF) for two more days, the excursion reduces to 0.3 dB. This confirms a reliable output of the light source and the need for more tolerant SMF coupling, as it is implemented in the following through the OHS to which the QKD transmitter of Alice couples over

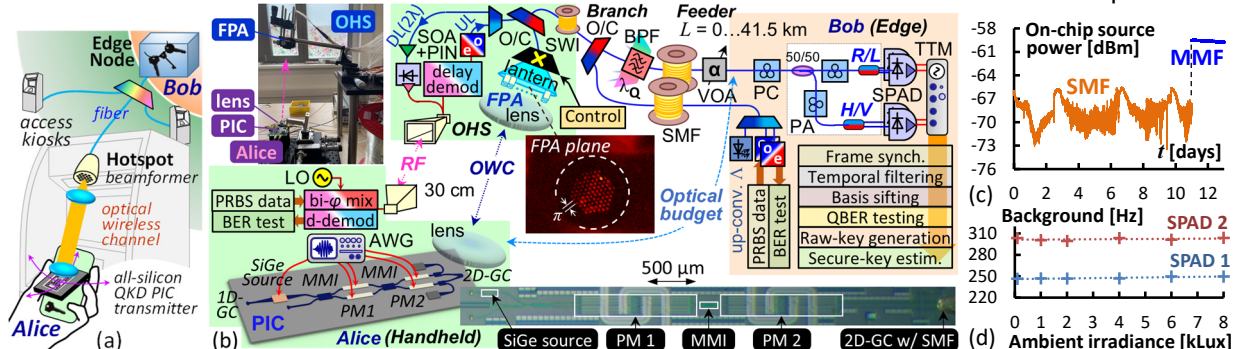


Fig. 1: (a) Fi-Wi enabled handheld QKD. (b) Experimental setup. The insets show the antenna elements of the FPA beamformer at the OHS and Alice's QKD transmitter PIC. (c) Stability of SiGe source emission. (d) Background upon ambient irradiance.

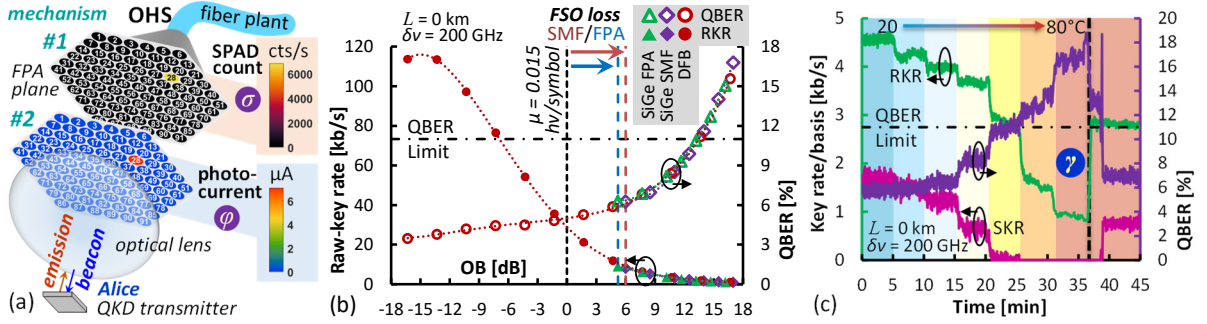


Fig. 2: (a) Finding Alice through the OHS' beamformer through SPAD-assisted signal detection (σ) or emitted beacon (φ). (b) QKD performance as a function of the OB. (c) Impact of operating temperature when pairing the QKD transmitter and OHS over air.

the close-proximity (30 cm) OWC link.

The OHS is comprised of an FPA beamformer with a space-switched beamforming network [18] to optimize the PIC-to-fiber coupling over air. The end facet of a photonic lantern (PL) with 91 fine-pitched ($\pi = 35 \mu\text{m}$) hexagonally arranged optical antenna elements compliant to the mode field of ITU-T G.652B fiber constitutes its focal plane (see inset in Fig. 1b). $\frac{1}{4}$ lenses with a focal length of 15.3 mm are employed at both sides of the OWC link. The total insertion loss of 1.6 dB for the FPA beamformer renders it suitable for QKD.

The OHS signal is then fronthauled to Bob's centralized QKD receiver through a WDM fiber distribution network. The WDM stage is emulated by a bandpass filter (BPF). Its bandwidth $\delta\nu$ slices Alice' LED-like SiGe source. Bob analyses the received polarization state in the H/V and R/L bases and InGaAs single-photon avalanche photodiodes (SPAD) with a detection efficiency of 8%, dark count rates of 248 and 302 Hz and a dead-time of 10 μs . After frame synchronization and basis sifting, the registered detection events are real-time processed upon temporal filtering to 50% of the symbol width to estimate the raw-key rate (RKR) and quantum bit error ratio (QBER). A variable optical attenuator (VOA) is inserted to evaluated these as a function of the optical budget (OB) between Alice and Bob.

Figure 2a demonstrates two mechanisms for alignment-tolerant coupling of Alice's quantum signal to the OHS, taking either Bob's registered SPAD counts (σ) or Alice's photocurrent (φ) upon reverse beacon injection as indicators. For the first, Alice's signal at $\lambda_Q = 1550.12 \text{ nm}$ is detected off the FPA center (element #28), showing a clear contrast between the peak of 6847 cts/s towards

other antenna elements. Given the launch after state preparation at Alice and fibre propagation, this corresponds to an optical coupling of -3.6 dB over the OWC channel. For the beacon-assisted mechanism we exploit the nature of Alice' silicon light source as dual-function device. By operating it as reversely biased photodetector while injecting a beacon of 1.1 dBm through the OHS, a photocurrent of 8.4 μA is generated for the same antenna element #28, yielding an equally good signal-to-noise ratio for localizing Alice within the field-of-view of the access terminal.

QKD Performance and Fi-Wi Co-Existence

We first report the QKD performance for single-feeder dark-fiber transmission without classical channels. Figure 2b introduces the back-to-back QKD baseline at 1550.12 nm for $L = 0 \text{ km}$. When operating the fully monolithic on-chip sourced QKD transmitter over the OWC link, a RKR of 9.1 (3.6) kb/s ($\blacktriangle$) at a QBER of 6.3% (8.2%) ($\triangle$) can be accomplished at an OB of 5.2 (10.2) dB. This translates to SKRs of 2.9 and 0.66 kb/s. An OB up to 13.2 dB can be supported before exceeding the QBER limit of 11% for secure-key extraction [19]. Comparison with single SMF coupling at the OHS ($\blacklozenge, \blacklozenge$) shows an improvement of 0.8 dB in OB. There is no penalty in RKR and QBER with respect to an externally sourced QKD transmitter with carved DFB source for light generation ($\bullet, \circ$). The restriction in OB when using the on-chip light source is due to its lower launch rate of $\mu = 0.015$ photons/symbol (for filtering at $\delta\nu = 200 \text{ GHz}$).

Figure 2c shows the coupling of the QKD transmitter PIC to an OHS employing a regular SMF under high operating chip temperature. Two effects are notable: First, loss of coupling without

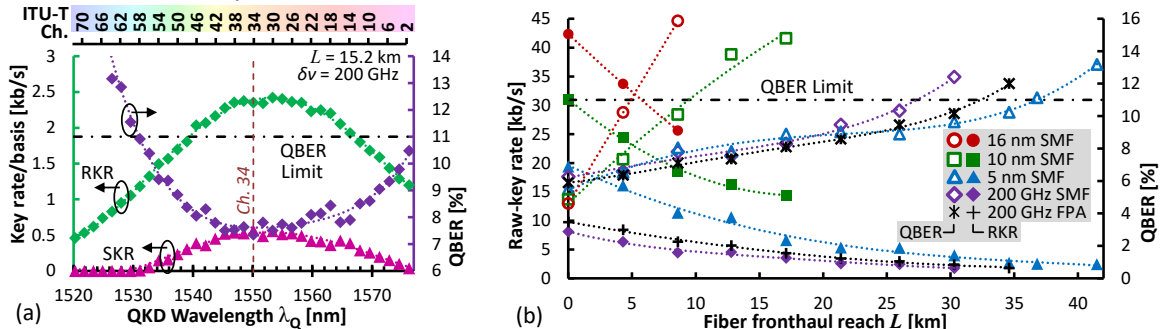


Fig. 3: (a) Wavelength-dependent QKD performance. (b) Evaluation of the QKD performance as a function of the fronthaul reach L and the slice bandwidth $\delta\nu$ for the QKD signal, for coupling to an OHS with either single SMF or the 91-element FPA beamformer.

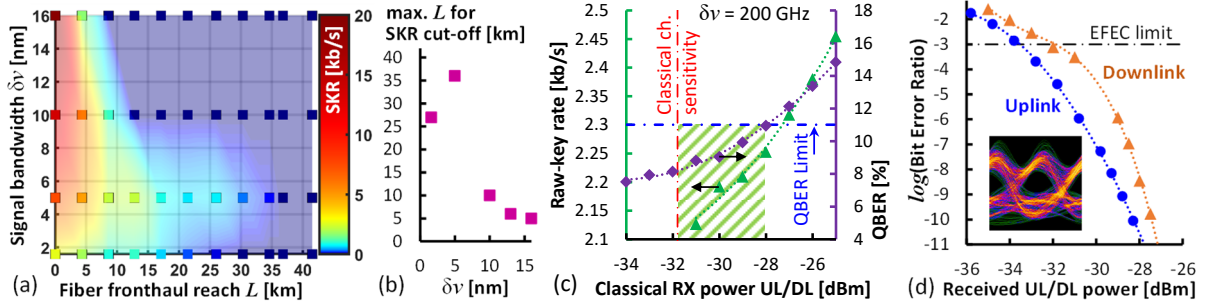


Fig. 4: (a) SKR estimation over fronthaul reach and optical slice bandwidth, and (b) fronthaul length at SKR cut-off. (c) QKD performance under co-existence with DL / UL radio-over-fiber transmission, and (d) classical BER for Fi-Wi radio transmission.

FPA beamformer causes a degradation of RKR and QBER. We can recover a QBER of 7.0% through manual re-alignment (γ). Second, the residual QBER penalty of 1.3% at 80°C results from the slightly degraded launch power of the QKD transmitter. Given the large margin towards the QBER limit, this is an acceptable trade-off in support of high-temperature operation.

Figure 3a shows the channel-dependent QKD performance over 30-cm OWC link and 15.2 km fiber reach. Secure key can be generated over 29 WDM (200-GHz) channels from 1531.1 to 1576.2 nm. 27 of these channels provide a sufficiently high SKR to secure 1 Tb/s connections according to the 256-bit AES-GCM limit for single-key use [20], assuming a standard nonce length of 96 bit (RFC5116), an Ethernet message length of 1500 MTU and an upper bound on attack probability of 2^{-60} (TLS 1.3). This renders the colorless QKD transmitter suitable for use in mobile networks with WDM fronthaul infrastructure and further supports programmable QKD schemes [21].

Figure 1d shows the influence of ambient light impinging on to the FPA aperture, for $\delta\nu = 200$ GHz. There is no increase in background noise beyond the SPAD darks for an irradiance up to 8000 lux. This confirms that the spatial filtering through the FPA antenna elements prevents detrimental QKD penalties and renders the proposed Fi-Wi link as suitable for key exchange for traditional in-door terminal installations.

Figure 3b reports the QKD performance at 1550.12 nm as function of the fiber reach for various optical slice bandwidths $\delta\nu$ from 200 GHz (1.6 nm) up to 16 nm. Depolarization [22] leads to reach limitations to 9.2 and 4.9 km for a large $\delta\nu$ of 10 nm ($\blacksquare, \square$) and 16 nm ($\bullet, \circ$), respectively – despite a higher RKR due to an increased launch μ of up to 0.1 $h\nu$ /symbol. On the other hand, the performance at $\delta\nu = 1.6$ nm ($\mu = 0.015$ $h\nu$ /sym) is similar for the OHS based on single SMF (reach limit of 27.2 km) and based on FPA beamformer, with the latter improving reach to 32 km by virtue of optimized coupling. There is no notable penalty due to depolarization for this $\delta\nu$ setting while for an intermediate $\delta\nu$ of 5 nm ($\blacktriangle, \triangle$) an excursion toward a reach of 36.5 km results from the interplay of an increased μ (and thus RKR), which over-compensates the depolarization-induced

QBER penalty. This is further confirmed by the estimated SKRs presented in Fig. 4a. Here, a clear SKR excursion is visible in the $\delta\nu$ region between 3 and 8 nm. The bandwidth-dependent fiber reach L at which the SKR cut-off is reached (Fig. 4b) indicates an optimum $\delta\nu$ around 4 nm, leading to a maximum reach of about 40 km.

Finally, we demonstrate co-existence with a FR3/mm-wave RF data shower between the kiosk and Alice. For this, downlink (DL) and uplink (UL) channels at 1305.98 and 1309 nm carry 5 Gb/s radio-over-fiber signals between the edge node and kiosk, with an additional CW wavelength at 1306.1 nm for photonic DL-BPSK up-conversion to 21 GHz. After over-the-air transmission using horn antennas, Alice applies delay demodulation to down-convert the RF data before bit error ratio (BER) estimation. Alice up-converts her 5 Gb/s uplink data to 27 GHz, using bi-phase modulation with a double-balanced RF mixer. Co-existence with QKD is further supported through O/C-band splitters and migration to a dual-feeder (14.3 km) dedicated to O- and C-band transmission, while retaining a joint branch segment of 512 m between WDM stage and kiosk (see Fig. 1b).

The QKD channel achieves a positive SKR for a bidirectional DL/UL launch of up to -18 dBm, which given the fronthaul loss of 10 dB relate to a received optical power (ROP) of -28 dBm (Fig. 4c). The BER for the DL ($\blacktriangle$) and UL ($\bullet$) is shown in Fig. 4d as a function of their ROP at kiosk and edge node, respectively. We obtain reception sensitivities of -27.4 (SOA+PIN) and -28.3 dBm (APD) at a BER of 10^{-10} . The SKR cut-off for the QKD channel shows a comfortable margin of 3.8 dB towards the EFEC limit (BER of 10^{-3}). Thus, coexistence is accomplished within the operating area marked as dashed green zone in Fig. 4c.

Conclusions

We have presented a hybrid Fi-Wi key exchange between an optical access terminal connecting Alice's QKD transmitter to a centralized Bob. Robust coupling over a beamformer-assisted close-proximity (30-cm) optical-wireless channel, operation at elevated temperatures of up to 80°C and positive SKRs in co-existence with a Fi-Wi radio signal confirm that a graceful introduction of simplified all-silicon QKD technology is feasible for applications in handheld commodities.

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